

System Blueprint (*a.k.a.* “Design Doc”)

TNPG: Restaurant Simulator

project: grocery simulator

Target ship date: 2026-05-25

Project Manager: Yu Lu

roster:

Name	Email	Primary Role	Secondary Role
Yu Lu	yul29@nycstudents.net	Front-end	Back-end
Ricky Lin	rickyl49@nycstudents.net	Middleware	Front-end
Jun Jie Li	junjiel26@nycstudents.net	Front-end	Art
Emily Mai	emilym384@nycstudents.net	Database Manager	Middleware

Summary

We plan to create a grocery store management simulator where the player takes on two roles: cashier and manager. During the day, customers enter the store, buy goods, and go to the register to pay, then the cashier will calculate the total cost and return the correct amount of change. During the night, the manager buys new stock for the next day and purchases store upgrades that improve customer mood and store efficiency.

Problem Being Solved

Improves user mental-math capabilities Improves user decision-making Improves user’s ability to plan for the future

Target Users

Who will use this system?

- Gamers interested in the simulator genres
- People interested in developing mental math skills

Why This Project Matters

Instead of practicing mental math through something like a worksheet, players can apply those skills in an interactive cashier system where they must calculate totals, handle money, and give correct change. Then in the manager role, players must decide how to spend their money, what stock to buy, and which upgrades are worth investing in for future days. These choices connect short term actions to long term outcomes, helping players practice resource management in a fun and low-pressure environment. We aim to improve the decision making and mental math skill of users through a fun and interactive format

Minimum Viable Product (MVP) Scope

Core Features (Required for Final Submission)

Features that must be completed:

1. Day - Cashier system
2. Customer generation
3. Grocery stock system

Stretch Features (Only if MVP is Complete)

1. Night - Manager system
2. Dialogue and Unique NPCS
3. Music and cutscenes

Explicit Non-Goals

Features intentionally excluded: - real-time multiplayer - Complex employee management systems (other than the player themselves) - character customization (except npc) - open-world - A fully randomized story mode —

Technology Stack

Layer	Selected Tool
Backend Framework	Flask
Frontend Framework	tailwind
Database	SQLite
Authentication	Flask sessions
ORM / DB Library	SQLAlchemy

Why This Stack Was Chosen

We considered our respective skillsets and the frameworks that we were comfortable with

Team Ownership Plan

Each member must own meaningful deliverables.

Team Member	Primary Ownership	Secondary Ownership	Specific Deliverables
Yu Lu	Templates, JS files	Flask routes	Core game loop and front-end
Jun Jie Li	Art assets	Templates, JS files	Art visuals
Ricky Lin	Flask routes	DB files, SQLITE tables	Flask routing between pages
Emily Mai	DB files, SQLITE tables	Flask Routes	Established database

Database Design

Component map

Site map

Key User Stories

1. As a player, I want to be as rich as possible so that I can beat the objective of the game
2. As a developer, I want to make the game interactive and functional so that we can fulfill the pleasures of our users
3. As a tester, I want to encounter and play through different games so that i can encounter scenarios with unintuitive/unexpected behaviors that developers might want to address

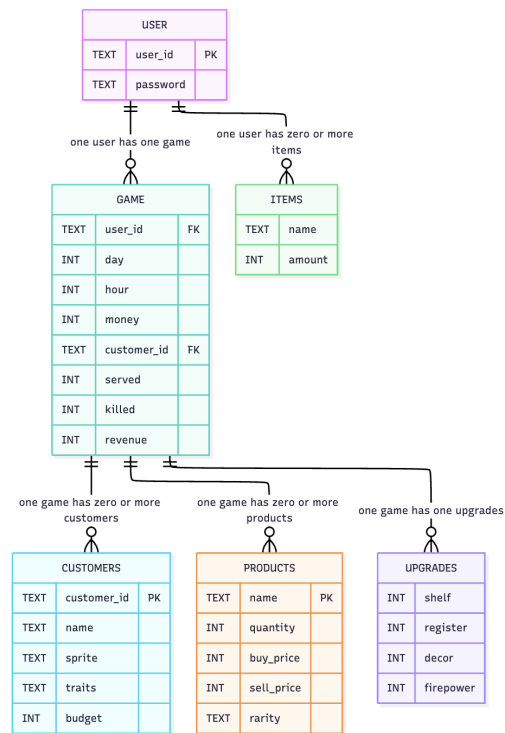


Figure 1: Database Algorithm

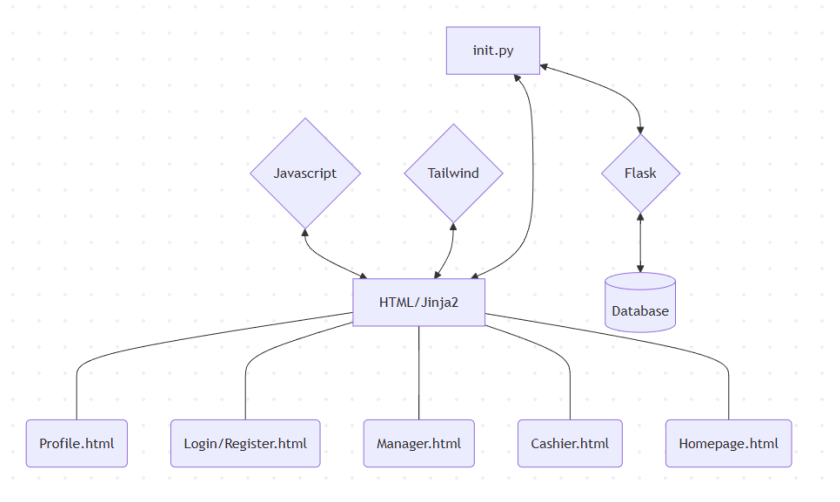


Figure 2: Component Map

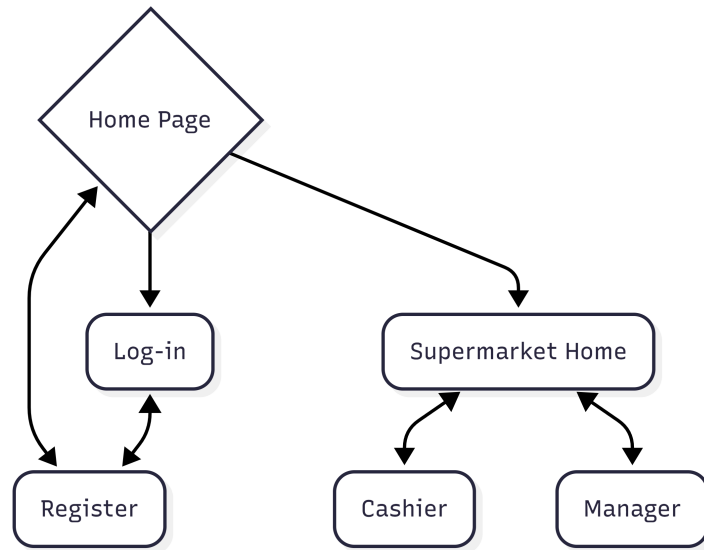


Figure 3: Grocery Site Map

Testing Plan

1. ALWAYS check developer console for errors and consoleLogs opening during testing
2. Play through 1 Day, serve customers and use the register and gun to see if they work as expected
3. Play through 1 night, buying upgrades and stock, and see if it is updated the next day
4. return to the website in another instance (or forced f5) to see if saves are working

Timeline

Week 1 Goals:

Design document, preliminary setup of repo, core game loop, save system

Week 2 Goals:

Basic art assets added, mainscreen done, user save saved to database and read/setup during refreshes

Week 3 Goals:

Cashier polishing:

1. Randomly generated customer appearances
2. Art for cashier terminal, corresponding art for background
3. chance for a misbehaving customer based on traits list

Night manager stock system:

1. Panel with ability to purchase stock/grocery for the next day
2. Implementation of a upgrades system that can affect gameplay
3. Art for background of manager Panel Storing/Retrieving of all of this into the database should be targeted

Internal Deadlines:

Special customers, game constants such as stock, prices, certain days, etc

Completion Criteria (*a.k.a.* “Definition of ‘Done’ ”)

Project is considered complete when all of the following are true:

1. Core game loop
2. Management system
3. Save system

Open Questions

1. should music/audio be added?
2. how many customer traits should affect gameplay, and how strong should their effects be?
3. should special NPCs appear on fixed days or randomly?
4. should customer purchases be fully random?

Appendix

Game Constants

- Total game length: 7 days
- Starting money: \$100
- Starting stock: apples, milk, pork
- Customers per day: 12
- Core player actions: calculate total, give change, buy stock, buy upgrades

Traits

- Robber: will leave with the food without paying, giving fake money
- Fat: buy much more food and money bonus

Upgrades

- Shelf upgrade: increases stock capacity
- Register upgrade: improves cashier efficiency
- Decor upgrade: improves customer mood
- Firepower: increases ammo count regeneration per day

Other

No additional notes